Water Quality Prediction in Arizona Using Gaussian Process Regression

CODE

Q1: Research Question

Problem:

- Dissolved metals (Fe, Cr, Mn, Mo, In) indicate water contamination risks
- Lab testing is costly ($50–$200/sample) and slow
- Phoenix metro has many untested water bodies

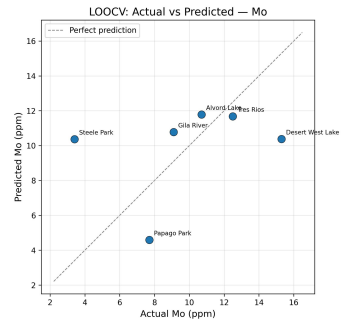
Goal:

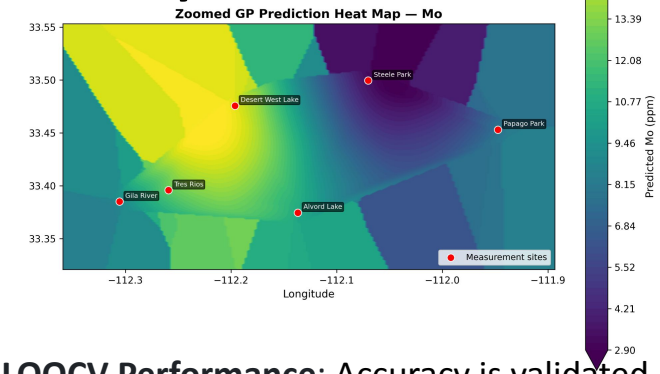
- Predict metal concentrations at unsampled locations
- Provide uncertainty estimates

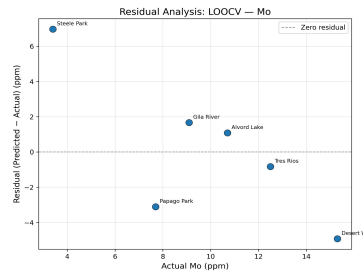
Q2: Project Design

- Water was collected from 6 Phoenix-area sites and filtered to remove particles, isolating only dissolved metals (Fe, Cr, Mn, Mo, In)
- Metal concentrations were measured using FAAS, a technique where each metal absorbs a unique wavelength of light; the more metal present, the more light is absorbed
- Concentration is calculated from absorbance using $A = \varepsilon \cdot l \cdot c$; a calibration curve of known standards converts raw readings into ppm values ($R^2 \geq 0.995$)
- A Gaussian Process (GP) is a statistical model that predicts metal concentrations at unmeasured locations based on patterns from the 6 sampled sites
- The model uses a kernel function to define how similar two locations are based on distance; closer sites are expected to have more similar concentrations

Q3: Data Analysis & Results







- **LOOCV Performance**: Accuracy is validated using Leave-One-Out Cross-Validation: the model is retrained on 5 sites and tested on the 6th, repeated for each site
- Predictive uncertainty increases **monotonically** with distance from training sites

Q4: Interpretation & Conclusions

Key Findings:

- Mn predicted most accurately. Fe showed the highest absolute error. Cr correctly ranked high-concentration western sites
- All five heatmaps share the same angular, patchwork boundary pattern between color zones
- Confidence was high near sampled sites and automatically dropped farther away

Advantages:

- Meaningful predictions from just 6 data points
- Every output includes a confidence range
- LOOCV confirmed the model generalizes to unseen sites; it was not overfitting to its 6 training points despite the small dataset

Applications & Next Steps:

- Use uncertainty maps to target future sampling and cut lab costs
- Expand to 15–20 sites and incorporate seasonal and land-use data